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Project Details

Project Title:	EfficientNet-based Cotton Disease Detection and Disease Treatment Support System for Smart Farming	
Type of Project	Software (AI-ML)	
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Abstract:

Cotton is a major agricultural crop, and its productivity is heavily affected by leaf diseases that often go unnoticed until significant damage has occurred. To address this issue, this project presents an ***EfficientNet-based Cotton Disease Detection and Treatment Support System*** *aimed at improving early diagnosis and decision-making in smart farming. The system uses a curated dataset of cotton leaf images and applies transfer learning on the EfficientNet model to accurately identify major diseases and nutrient deficiencies. Along with classification, the system estimates the severity of the infection, which helps farmers understand the stage and extent of crop damage. Based on the detected condition, the system provides treatment recommendations that include both chemical pesticides and eco-friendly organic alternatives such as neem oil, bio-fungicides, and micronutrient sprays. The solution is designed for deployment on a mobile or web platform, enabling farmers to obtain real-time disease diagnosis simply by uploading a leaf image. By combining deep learning with practical field recommendations, this project aims to support timely treatment, reduce crop losses, and promote more sustainable cotton farming practices.*

1. Introduction

Cotton is one of the most important agricultural crops in India and worldwide, yet its productivity is often reduced by various leaf diseases and pest attacks that frequently go unnoticed until significant damage has occurred. Common issues such as **Aphids**, **Army worm**, **Bacterial Blight**, **Powdery Mildew**, and **Target Spot**, along with nutrient deficiencies, weaken plant growth, lower yield, and reduce fiber quality, ultimately affecting the income and stability of cotton-growing farmers. Traditional disease detection methods usually depend on visual inspection or local guidance, which may lead to misdiagnosis, delayed treatment, and unnecessary pesticide use, contributing to increased production costs and long-term soil damage. To overcome these challenges, this project introduces **an EfficientNet-based Cotton Disease Detection and Treatment Support System** aimed at enabling early, accurate, and accessible diagnosis.

The system is built using a curated dataset of cotton leaf images representing major diseases—including Aphids, Armyworm, Bacterial Blight, Powdery Mildew, Target Spot—and healthy leaves. By applying transfer learning on the EfficientNet architecture, the model can effectively identify diseases and nutrient deficiencies while maintaining high computational efficiency, making it suitable for real-world deployment. Beyond classification, the system includes a severity estimation module that categorizes infections as mild, moderate, or severe, helping farmers understand the extent and urgency of the problem. Based on the detected disease and its severity, the system provides tailored treatment recommendations covering both chemical pesticides and eco-friendly organic solutions like neem oil, bio-fungicides, micronutrient sprays, and soil amendments.

Designed for mobile and web platforms, the solution allows farmers to simply upload a leaf image and receive instant diagnostic results, reducing dependency on experts and ensuring timely intervention. By combining deep learning with practical treatment guidance, this project aims to reduce crop losses, support sustainable agricultural practices, and enhance decision-making in cotton farming. Overall, the system demonstrates how modern AI technologies can be integrated into agriculture to improve plant health management, promote eco-friendly farming, and strengthen the livelihoods of cotton cultivators.

In addition, the project lays the foundation for future enhancements such as real-time field monitoring, multi-disease detection, and integration with IoT-based crop advisory systems. With further development, it can be extended to detect stem and boll diseases, incorporate climatic data, and provide region-specific recommendations to farmers. This approach aligns with the growing vision of digital agriculture, where intelligent tools assist farmers in making data-driven decisions and improving overall productivity. Overall, this project highlights the potential of modern AI technologies to address real-world agricultural challenges. By combining deep learning with user-friendly interfaces, it demonstrates how intelligent systems can move from research labs to actual farm fields, offering meaningful support to farmers and improving the overall efficiency of cotton cultivation.

2. Literature Survey:

Author / Title	Publications	Methodology	Merits	Research Gaps
Eshika Jain, Danish Kundra Automated Detection & Classification of Cotton Leaf Diseases Using DenseNet201	ICOCT, 2025	DenseNet201 model trained on 875 cotton leaf images with preprocessing & augmentation.	94% accuracy, strong feature extraction, good performance.	Small dataset, confusion among similar diseases, needs field-based validation.
Swastick Ganguly et al. Automated Detection & Classification of Cotton Leaf Diseases	AMATHE, 2024	Hybrid CNN model combining DenseNet121 + ResNet50 for 6 disease classes.	High 96% accuracy, strong classification capability.	High model complexity, limited real-world testing, dataset not diverse.
Samarth Otiya et al. Cotton Leaf Disease Classification Using Deep CNN with XAI	ICSCNA, 2023	Used transfer learning (ResNet50, Xception, EfficientNet); explainability via Grad-CAM.	Very high accuracy (up to 99%), good model transparency.	Small dataset, class imbalance, weak field generalization.
Ruthvija Kesani et al. Cotton Plant Disease Detection & Prevention	ICRITO, 2024	CNN with object detection; drone-based monitoring for real-time detection.	Real-time capability, good accuracy across models.	Needs larger datasets, drone cost, limited geographic scalability.
Sandhya Dhage, Vijay Kumar Garg Cotton Plant Fungal Disease Classification	ICETET-SIP, 2023	Compared 18 DL models using real-field fungal disease images.	High accuracy, strong feature extraction.	Dataset imbalance, limited fungal variety, needs deployment system.
Soumyajit Bera, Dr. Puneet Kumar Deep Learning Approach in Cotton Disease Detection	INOCON, 2023	Survey of CNN, TL, GAN, SVM and other DL methods.	Comprehensive overview, identifies best-performing models.	No experimental validation, lacks real-world data insights.

3. Positioning

3.1 Problem statement:

EfficientNet-based Cotton Disease Detection and Disease Treatment Support System for Smart Farming

3.2 Product position statement:

The proposed **EfficientNet-based Cotton Disease Detection and Treatment Support System** is designed for farmers, agricultural advisors, and agri-tech organizations who require fast and reliable plant disease identification. Our product provides an AI-driven platform that can detect cotton leaf diseases, estimate severity, and recommend suitable chemical and organic treatments. Unlike traditional manual diagnosis or general-purpose farming apps, this system offers disease-specific recommendations powered by deep learning, helping users make timely, informed, and sustainable decisions to protect crop health.

4. Planning

The planning of the **EfficientNet-based Cotton Disease Detection and Treatment Support System** is organized into simple, clear stages to ensure smooth project execution. The focus is on understanding the problem, preparing the dataset, developing the model, integrating treatment recommendations, and finally testing the system in real scenarios.

Key Planning Steps

1. **Understanding the Problem**
   - Study common cotton leaf diseases and nutrient deficiencies.
   - Review existing AI-based plant disease detection solutions.
2. **Dataset Preparation**
   - Collect cotton leaf images from reliable sources.
   - Organize, preprocess, and augment the data for model training.
3. **Model Development**
   - Use EfficientNet with transfer learning to classify diseases.
   - Train, fine-tune, and evaluate the model using appropriate metrics.
4. **Severity Estimation**
   - Design a simple mechanism to determine whether a disease is mild, moderate, or severe.
5. **Treatment Recommendation**
   - Prepare disease-specific suggestions, including chemical and organic treatments.
   - Link recommendations with the severity level.
6. **System Integration & UI**
   - Connect the trained model with a backend API.
   - Develop an easy-to-use interface for image upload and results.
7. **Testing and Review**
   - Validate the system with test images.
   - Fix issues, improve performance, and finalize documentation.

5. Project Scope

The scope of this project is centered on designing and developing an intelligent, AI-based system capable of accurately detecting cotton leaf diseases and providing practical treatment recommendations to support farmers and agricultural stakeholders. The primary focus is on building a deep learning model using the EfficientNet architecture, which can classify major cotton leaf diseases, identify nutrient deficiencies, and estimate the severity level of the infection. By integrating image-based diagnosis with treatment suggestions, the system aims to serve as a comprehensive support tool for smart farming.

The project scope covers the complete end-to-end workflow of model development. This includes dataset collection from farms and public repositories, image preprocessing, data augmentation, model training, validation, and performance evaluation. The system is further enhanced with a severity-estimation module that analyzes the extent of infection and categorizes it as mild, moderate, or severe, assisting farmers in choosing the correct dosage and treatment method. Alongside this, the treatment support module provides both chemical and organic solutions, helping reduce dependency on guesswork and minimizing excessive pesticide usage.

In addition to model development, the scope also includes creating a user-friendly interface in the form of a web or mobile application. Through this interface, users can upload cotton leaf images and receive instant results regarding disease type, severity, and recommended treatment plans. This makes the system practical and accessible for field use, even for farmers with minimal technical knowledge. The prototype aims to operate efficiently on low-resource devices and provide results in real time.

While the system is designed to be highly functional, the current scope is limited to leaf-level disease detection. It does not cover stem, boll, or soil-related diseases. Similarly, the model is trained only on diseases present in the curated dataset and may require future expansion to include rare or region-specific conditions. The project does not include farm-level yield prediction or climatic risk modeling, as the main focus remains on disease diagnosis and treatment support.

Despite these boundaries, the project provides a strong foundation for automating disease detection in cotton farming. It demonstrates how modern AI techniques can assist farmers in making faster and more accurate decisions, reduce crop losses, and promote sustainable agricultural practices. The system developed through this project can also be expanded, scaled, and adopted by agricultural institutions, agritech companies, and government departments working toward digital agriculture and improved rural livelihood.

6. Methodology

The methodology used for developing the **EfficientNet-based Cotton Disease Detection and Treatment Support System** follows a structured, step-by-step workflow that ensures reliable disease detection, accurate severity estimation, and practical treatment suggestions. Each stage contributes to building a robust and farmer-friendly solution.

1.1 Dataset Collection

Cotton leaf images are collected from publicly available datasets and verified agricultural sources. The dataset contains healthy leaves, multiple disease categories, and nutrient deficiency samples to cover a wide range of real-field conditions. To maintain fairness in training, the images are split into **training**, **validation**, and **testing** sets in an 80:10:10 ratio.

6.2 Data Preprocessing

Before training, all images undergo preprocessing to make them suitable for the model. Steps include:

- Resizing images to **224 × 224 pixels**
- Normalizing pixel values
- Applying augmentation (rotation, zoom, cropping, brightness change, flipping)

These steps increase dataset variety, reduce overfitting, and help the model generalize better to real-world farm images.

6.3 Model Training Using EfficientNet

The EfficientNet architecture, pretrained on ImageNet, is fine-tuned using **transfer learning**. A custom classification layer is added to classify cotton-specific diseases. The training process uses:

- Adam optimizer
- Early stopping to prevent overfitting
- Learning rate scheduling for stability

Model performance is monitored using metrics such as **accuracy, precision, recall, F1-score,** and **confusion matrix**.

6.4 Disease and Deficiency Detection

Once the model is trained, it predicts the disease category for any cotton leaf image uploaded by the user. It generates:

- The predicted class
- A confidence score

The system can identify common conditions like **bacterial blight, leaf spot, curl virus**, and **nutrient deficiencies**, along with detecting healthy leaves.

6.5 Severity Estimation

To help farmers understand how serious the disease is, a severity estimation module examines:

- Infection spread
- Lesion intensity
- Affected area

6.6 Treatment Recommendation System

After identifying the disease and its severity, the system provides practical treatment suggestions. These include:

- **Chemical solutions:** pesticides or fungicides with recommended dosage
- **Organic options:** neem oil, bio-fungicides, micronutrient sprays
- **Preventive measures:** cultural practices to avoid recurrence

This makes the system helpful for both conventional and eco-friendly farming approaches.

6.7 System Deployment

The final model is integrated into a simple and intuitive web or mobile interface. Users can upload an image of a cotton leaf, and the system instantly displays:

- Disease name
- Severity level
- Recommended treatments

This deployment ensures that farmers and agricultural advisors can access real-time guidance directly from the field.

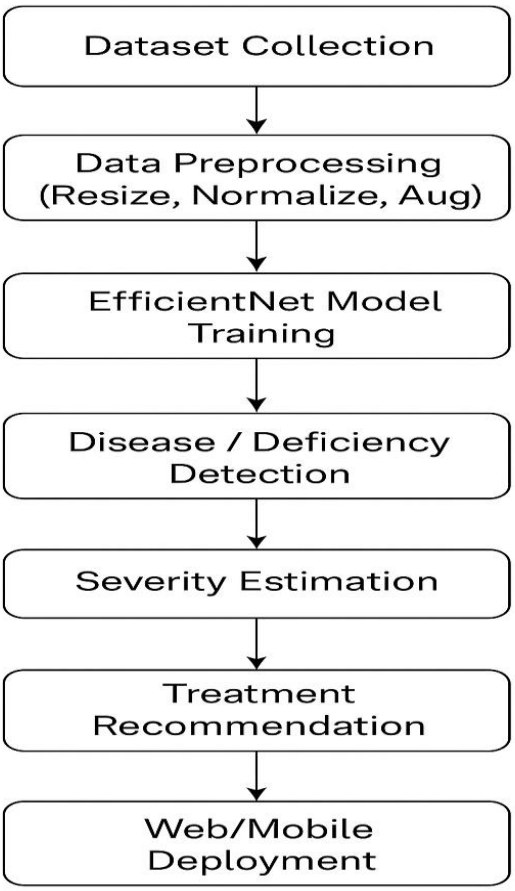

Fig.6.71 System Architecture

7. Modules Identified

The proposed EfficientNet-based Cotton Disease Detection and Disease Treatment Support System is divided into several interconnected modules. Each module performs a specific function that contributes to the overall workflow of disease detection, severity estimation, and treatment recommendation.

7.1 Dataset Collection Module

This module gathers cotton leaf images from public datasets and field sources. It ensures the availability of diverse samples representing healthy leaves, major diseases, and nutrient deficiencies. The dataset is properly labeled and organized for efficient training and evaluation.

7.2 Data Preprocessing Module

This module handles image preparation tasks, including resizing, normalization, and augmentation (rotation, cropping, brightness adjustment, and flipping). These operations improve model robustness and reduce the chances of overfitting.

7.3 EfficientNet Classification Module

This is the core deep-learning component of the system. The EfficientNet model uses transfer learning to classify the input cotton leaf image into predefined disease or deficiency categories. It outputs the predicted class along with a confidence score.

7.4 Severity Estimation Module

Once a disease is detected, this module analyzes the infection area and lesion intensity to determine the severity level—mild, moderate, or severe. The output helps farmers understand how urgent the treatment requirement is.

7.5 Treatment Recommendation Module

Based on the predicted disease and severity level, this module generates appropriate treatment suggestions. It includes:

Chemical pesticide/fungicide recommendations with dosage,

Organic and eco-friendly solutions like neem oil, bio-fungicides, and micronutrient sprays,

Preventive practices to reduce future outbreaks.

7.6 User Interface & Deployment Module

This module integrates the trained model into a user-friendly web/mobile platform. It enables users to upload leaf images and receive instant disease diagnosis, severity details, and treatment recommendations. The UI ensures accessibility for farmers and agricultural advisors.

8. Requirement and cost Analysis

The successful development of the EfficientNet-based Cotton Disease Detection and Disease Treatment Support System requires a combination of hardware, software, human resources, and operational tools. This section outlines all necessary requirements and provides an estimated cost analysis for implementing and testing the project.

8.1 Hardware Requirements

Hardware Component	Specification	Purpose
Laptop/PC	Intel i5/i7 or Ryzen 5/7, 8–16 GB RAM	Model training, coding, testing
GPU (Optional but preferred)	NVIDIA GTX 1650 / RTX Series	Faster deep learning model training
Smartphone/Web-enabled device	Android/iOS	Testing mobile/web deployment
Internet Connectivity	Broadband/WiFi	Dataset download, cloud services

8.2 Software Requirements

Software / Tool	Version	Purpose
Python	3.8+	Model development
TensorFlow / Keras	Latest	EfficientNet model training
OpenCV	Latest	Image preprocessing
Kaggle Notebook / VS Code	---	Coding environment
HTML/CSS/JS	---	User interface components
Kaggle Dataset	---	Cotton leaf dataset
GitHub	---	Code repository

8.3 Human Resource Requirements

Role	Responsibility
Project Lead	Overall coordination, task assignment
ML Developer	Model development, training, evaluation
Data Preprocessing Engineer	Dataset cleaning, augmentation
UI/UX Developer	Front-end interface design
Backend Developer	Server integration, API creation
Documentation Lead	Report writing, formatting, presentation

8.4 Cost Estimation

A. Hardware & Infrastructure Cost

Item	Estimated Cost
Laptop/PC (already available for most students)	₹0 – ₹50,000
Cloud GPU (optional, for faster training)	₹3000 – ₹3,500
Internet Charges	₹1000 – ₹1,500

B. Software Cost

Most software tools used (Python, TensorFlow, GitHub, Flask, Streamlit) are open-source and free.

Total Software Cost: ₹0

C. Operational Cost

Item	Estimated Cost
Dataset collection (internet, storage)	₹300 – ₹500
Miscellaneous (printing, documentation)	₹200 – ₹400

8.5 Total Estimated Project Cost

Category	Estimated Range
Hardware & Cloud	₹3,800 – ₹5,200
Software Tools	Free
Documentation & Misc	₹2,600 – ₹3,500
Total Estimated Cost	₹10,000 (approx.)

The project is cost-effective as most resources are open-source. The main expenses arise from optional GPU usage, internet charges, and minor documentation costs. This makes the system feasible for academic research as well as practical implementation in agri-tech environments.

9. Developing project execution timelines

The development of the EfficientNet-based Cotton Disease Detection and Treatment Support System is planned across multiple phases to ensure structured progress and timely completion. The overall timeline spans 20–24 weeks, covering research, model development, testing, and deployment.

Week 1–4: Research & Planning

- Study cotton diseases and related research
- Review EfficientNet architecture
- Finalize problem statement and system design

Week 5-6: Dataset Collection

- Collect cotton leaf images
- Organize and label dataset

Week 7–9: Data Preprocessing

- Resize, normalize, and augment images
- Split dataset into train/validation/test sets

Week 9–12: Model Training

- Fine-tune EfficientNet using transfer learning
- Evaluate metrics and improve accuracy

Week 13-15: Severity Estimation

- Implement severity calculation
- Categorize output into mild/moderate/severe

Week 16-18: Treatment Recommendation

- Create treatment rule base
- Add chemical + organic suggestions

Week 19–21: Integration & UI Development

- Build backend using Flask/Streamlit
- Develop simple web/mobile UI for image upload

Week 22--24: Final Testing & Documentation

- Test complete system
- Fix bugs and prepare final project report

10. Survey Paper Publication details

As part of the project documentation and research activities, the team has conducted an extensive survey on existing deep learning approaches, cotton disease datasets, EfficientNet-based classification techniques, severity estimation methods, and treatment recommendation systems used in precision agriculture. The survey work is aligned with the project objectives specified in the **KSCST(Karnataka State Council for Science and Technology)** proposal, which focuses on smart farming, rural livelihood improvement, and sustainable agriculture.

This paper reviews:

- Major cotton leaf diseases and nutrient deficiencies
- Dataset sources such as the Kaggle Cotton Plant Disease dataset
- Image preprocessing and augmentation techniques
- Transfer learning with EfficientNet and comparable CNN architectures
- Severity estimation approaches
- Chemical, organic, and eco-friendly treatment strategies
- Gaps in existing systems and the motivation for the proposed model

The survey paper is currently in its pre-publication stage and is being prepared for submission in alignment with academic and research guidelines. As part of the KSCST Student Project Programme (SPP) – 49th Series, the work will also be presented at the **KSCST Annual State-Level Exhibition (Poster Presentation)**. The final version of the paper will include detailed performance metrics, model evaluation results, and an analysis of the treatment recommendation system, ensuring that the publication reflects the complete technical outcomes of the project.

11. Conclusions

The EfficientNet-based Cotton Disease Detection and Treatment Support System highlights the effectiveness of deep learning in modern agricultural problem-solving. By leveraging a fine-tuned EfficientNet model and a carefully curated dataset, the system is able to detect various cotton diseases and nutrient deficiencies with promising accuracy. Its ability to analyze leaf images, estimate disease severity, and provide scientifically backed treatment suggestions makes it a practical and farmer-friendly solution. The inclusion of both organic and chemical treatment options broadens its usability, allowing different categories of farmers—small-scale, organic growers, and commercial producers—to benefit from the system. This reduces dependence on manual inspection, mitigates the risk of misdiagnosis, and ensures timely interventions that are crucial for protecting crop health.

In a broader context, the project demonstrates how AI-powered tools can support precision agriculture by reducing diagnostic time, minimizing crop loss, and improving decision-making at the farm level. Such systems can lead to more sustainable farming by preventing excessive pesticide usage and promoting targeted treatments based on disease severity. Looking ahead, the system has strong potential for expansion through features such as mobile deployment, real-time image capture, multilingual farmer interfaces, and integration with weather and soil data for predictive analysis. With further development and larger datasets, this solution can evolve into a comprehensive smart farming platform capable of supporting multiple crops and providing end-to-end guidance for plant health management.

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